NATURAL LANGUAGE PROCESSING PROJECT

Twitter Sentiment Analysis Classifier for Apple and Google Products

Using Natural Language Processing (NLP) to classify Twitter sentiment
as Positive, Negative, or Neutral toward Apple and Google products

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AGENDA

What We'll Cover

1. Business Understanding
2. Data Understanding
3. Data Preparation & Cleaning
4. Exploratory Data Analysis
5. Building the Models
6. Model Evaluation & Selection
7. Model Interpretation
8. LIME Explanations
9. Key Findings
10. Recommendations & Conclusion

01 · BUSINESS UNDERSTANDING

The Problem

- Thousands of consumers share opinions about Apple and Google products on Twitter.
- Manually monitoring these opinions is difficult and time-consuming for companies.
- An automated NLP model is needed to classify tweets as Positive, Negative, or Neutral.
- This supports brand monitoring, customer satisfaction tracking, and faster decision-making.

GOAL

Build a model that automatically classifies tweet sentiment toward Apple and Google products into Positive, Negative, or Neutral categories — helping marketing, customer experience, and brand teams understand public opinion at scale.

Stakeholders & Key Questions

Marketing Team

Shape campaigns based on public opinion

Customer Excellence

Spot complaints & recommend fixes

Product & Brand Managers

Identify brand strengths & weaknesses

Company Executives

Support strategic decision-making

KEY BUSINESS QUESTIONS

- Can tweets be automatically classified as positive, negative or neutral?
- What is the overall sentiment toward Apple and Google products?
- Which products generate the most positive/negative reactions?
- What words are associated with each sentiment?
- How accurately can ML predict tweet sentiment?

Dataset Composition

Sourced from CrowdFlower via data.world. The dataset contains tweets about Apple and Google products with sentiment labels.

9,093
Total Tweets

3
Key Columns

2
Brands Covered

KEY VARIABLES

- tweet_text — the original tweet content
- sentiment — Positive, Negative, Neutral, or Can't Tell
- target_brand — the Apple or Google product/brand mentioned

TARGET VARIABLE: SENTIMENT DISTRIBUTION

Sentiment	Count	%
No emotion	5,389	59.3
Positive	2,978	32.8
Negative	570	6.3
Can't tell	156	1.7
Total	**9,093**	**100.0**

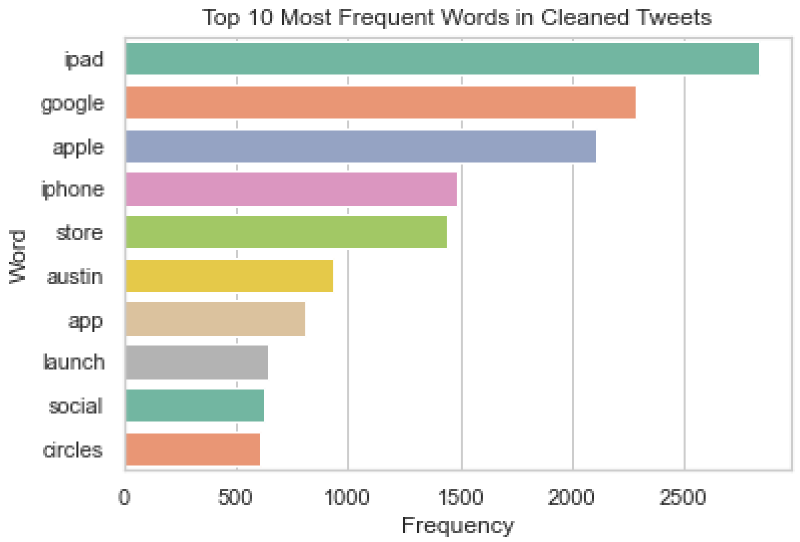
The dataset is highly imbalanced — "No emotion" dominates at 59%, while negative sentiment is only 6.3%. This makes F1-score more reliable than accuracy.

Cleaning & Preprocessing the Tweets

TEXT PREPROCESSING STEPS

1. Convert text to lowercase
2. Remove URLs and links
3. Remove Twitter handles & hashtags
4. Remove punctuation & special characters
5. Remove stopwords
6. Lemmatization

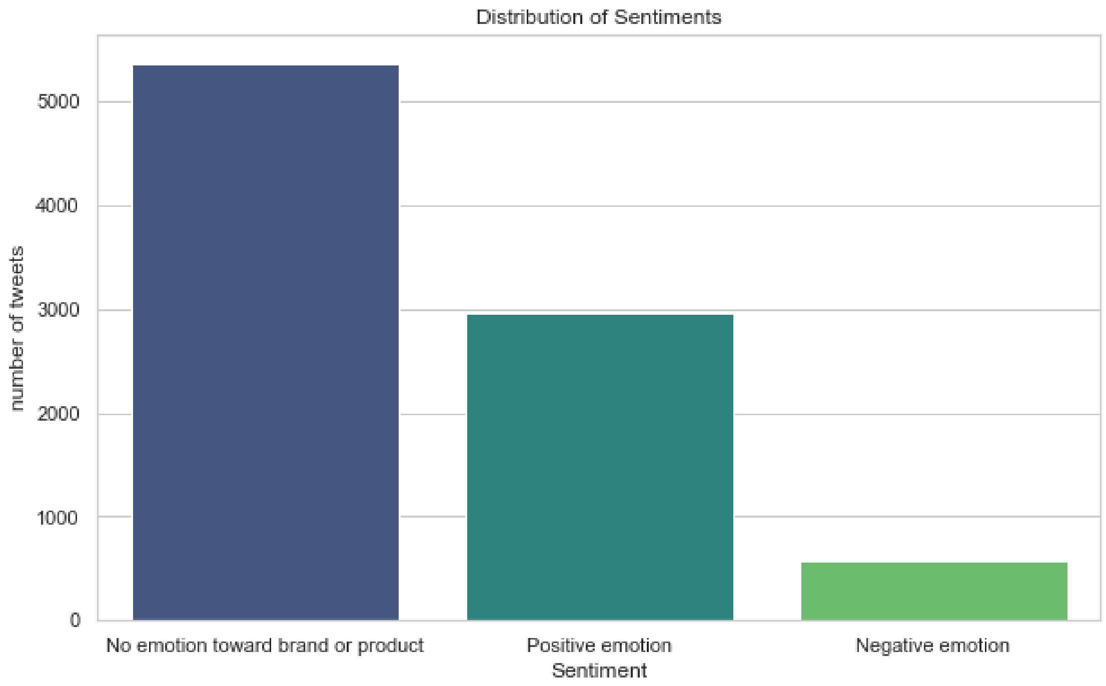
TOP 10 MOST FREQUENT WORDS (CLEANED)

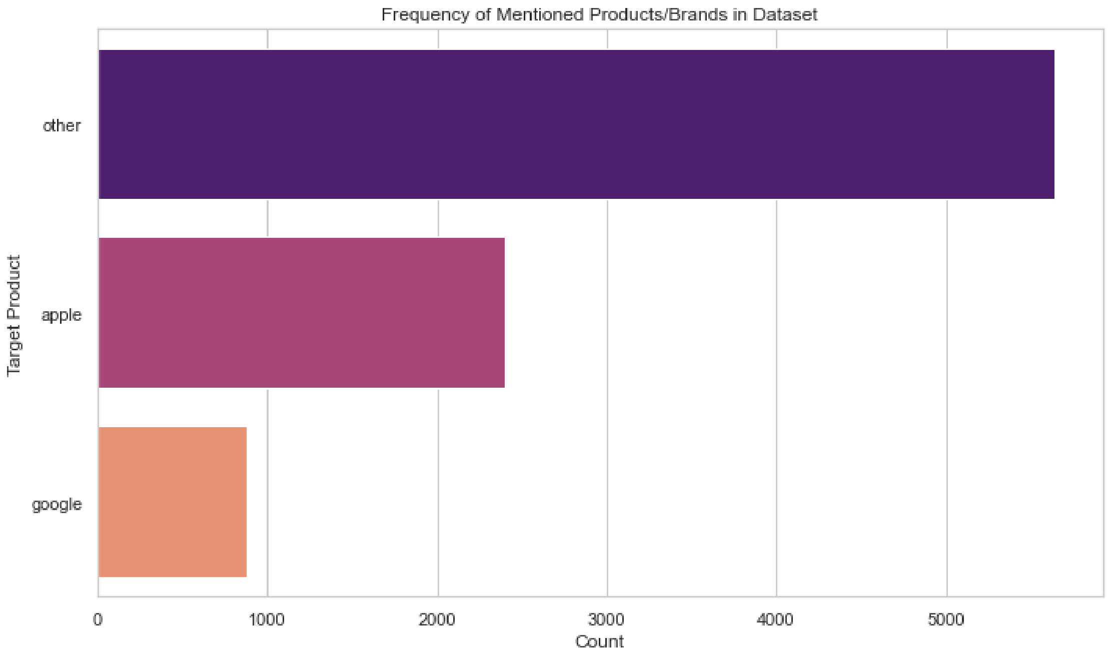


DATA CLEANING PROCESS

- Removed missing/null tweet rows and duplicates
- Dropped tweets labelled "I can't tell"
- Simplified target_brand into Apple, Google, and Other

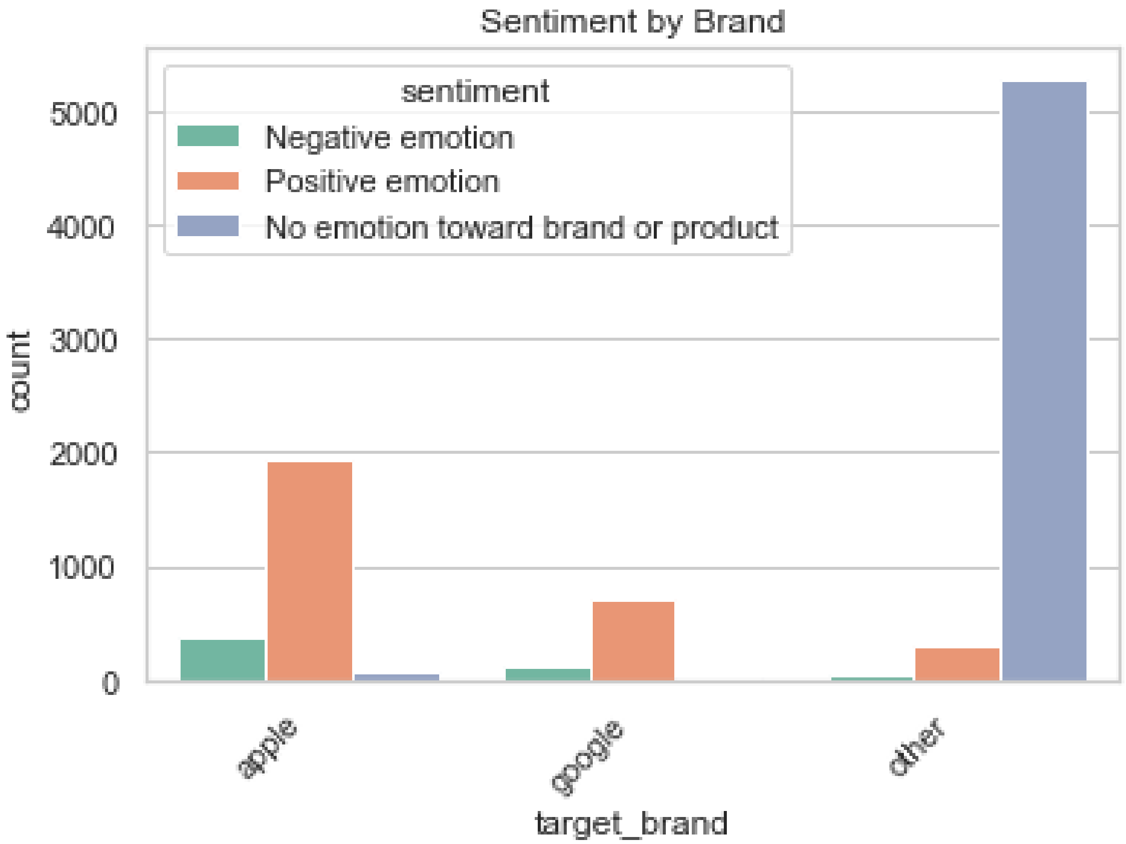
Sentiment & Brand Distribution

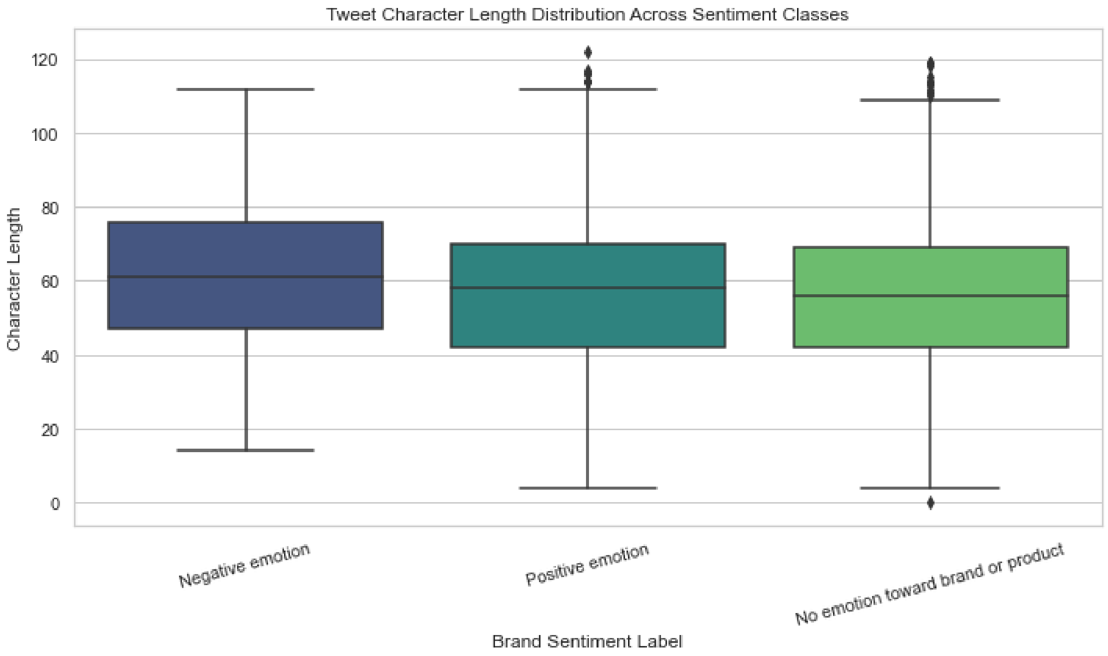




Most tweets are neutral (no clear emotion), followed by positive sentiment. Negative sentiment is the smallest category. Apple-related products dominate brand mentions over Google.

Sentiment by Brand & Tweet Length





Sentiment distribution differs by brand — positive tweets generally outnumber negative ones. Tweet character length varies modestly across sentiment classes, with neutral tweets often slightly longer.

Models Explored

Tweets were converted to numeric features using TF-IDF (unigrams + bigrams). Three classification algorithms were trained and compared on the multiclass sentiment task.

Model 1

Logistic Regression

Baseline model — strong on high-dimensional sparse text data and easily interpretable. Tested in binary and multiclass form, with class-weight balancing applied to address imbalance.

90% accuracy (multiclass)

Model 2

Multinomial Naive Bayes

Fast and simple, works well with word-frequency features. Struggled significantly with the minority negative class.

Low negative recall (3%)

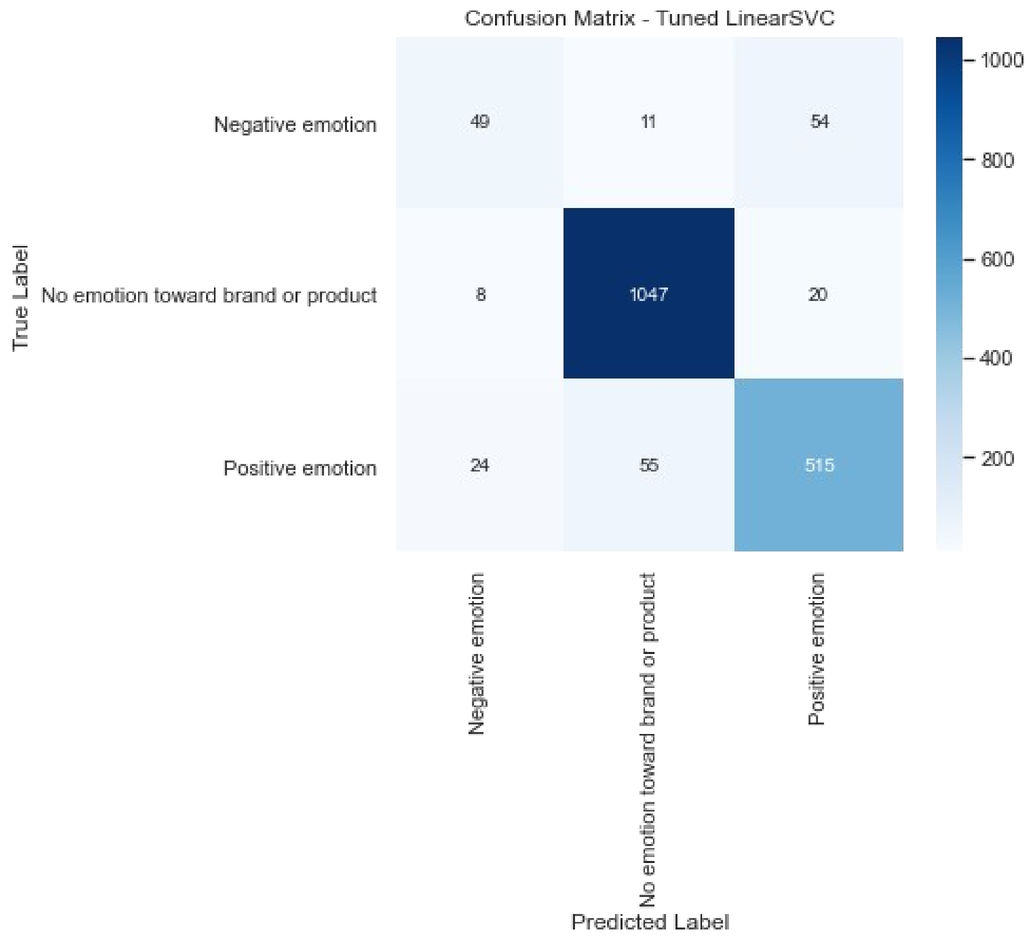
Model 3 BEST MODEL

LinearSVC (Tuned)

Effective for high-dimensional text data. Tuned via GridSearchCV with class-balancing — selected as the best-performing model.

90.3% accuracy, Macro F1 ≈ 0.76-0.78

Best Model: Tuned LinearSVC



MODEL SUMMARY

Metric	Value
Final Model	LinearSVC (tuned with GridSearchCV)
Test Accuracy	90.3%
Macro F1-Score	~0.76 - 0.78
Tuning Method	GridSearchCV (3-fold CV)
Evaluation Metric	Macro F1-Score

The confusion matrix shows the model is highly effective at identifying neutral and positive sentiments. Negative sentiment remains the hardest class to predict, though it improved noticeably after tuning and class-weight balancing.

07 · MODEL INTERPRETATION

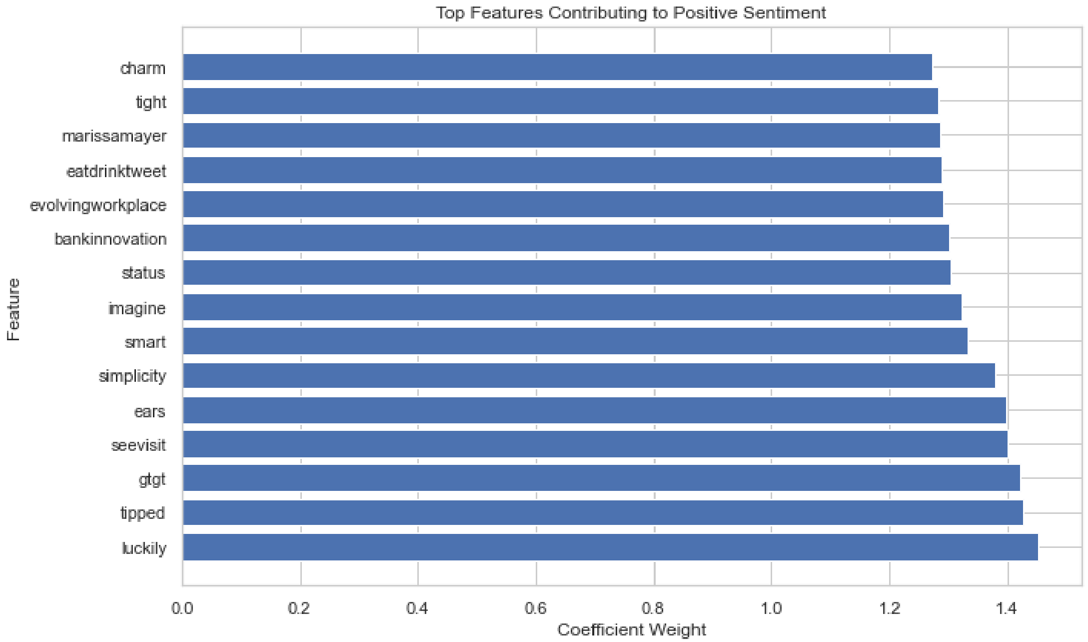
What Drives the Predictions?

Since LinearSVC is a linear model, each word/feature has a coefficient (weight) showing how strongly it pushes a prediction toward a sentiment class. Larger positive weights mean stronger influence on that class.

HOW IT WORKS

- TF-IDF + one-hot encoded brand features feed the model
- Each class (Positive, Negative, Neutral) has its own coefficient set
- Words with the highest positive weights for a class are its strongest predictors
- This makes the model's decisions transparent and explainable to stakeholders

TOP FEATURES — POSITIVE SENTIMENT CLASS



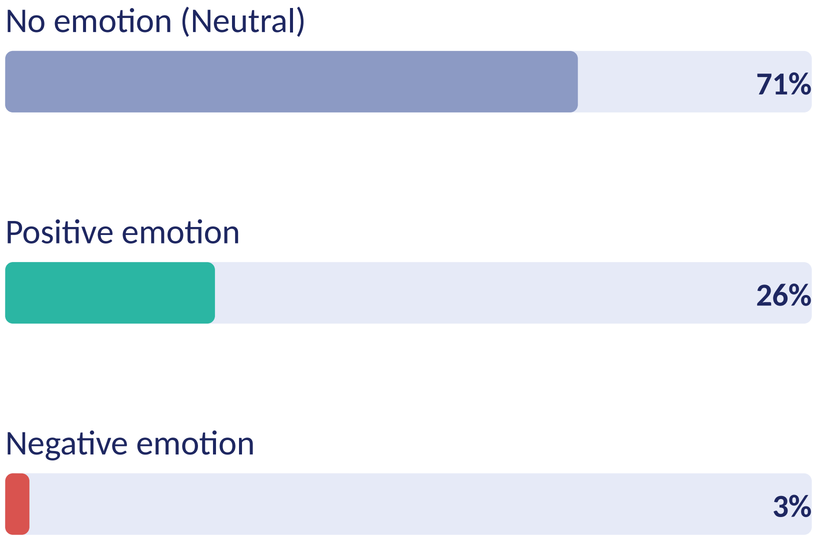
Explaining Individual Predictions with LIME

LIME (Local Interpretable Model-Agnostic Explanations) approximates the model locally around a single tweet, highlighting which words pushed the prediction toward each sentiment class.

EXAMPLE TWEET EXPLANATION

- Predicted class: "No emotion toward brand or product" (probability 0.71)
- Other probabilities: 0.26 toward Positive, 0.03 toward Negative
- Words like "longest" and "line" pushed the prediction strongly toward Neutral
- The word "far" contributed slightly toward a Negative interpretation, reflecting dissatisfaction

CLASS PROBABILITIES



LIME helps explain WHY a single tweet was classified a certain way — not just the overall model behaviour.

What We Learned

1

Neutral sentiment dominates

Most tweets express the brand without a strong emotional tone, suggesting general/informational discussion.

2

Negative sentiment is rare but important

Only ~6% of tweets are negative — the minority class is hardest for models to detect.

3

Class imbalance affects performance

Models lean toward predicting the majority neutral class unless balancing techniques are applied.

4

Preprocessing improved data quality

Cleaning (lowercasing, removing URLs/handles/stopwords, lemmatization) reduced noise significantly.

5

Simple models struggle with minority classes

Logistic Regression and Naive Bayes had high overall accuracy but poor recall on negative tweets.

6

LinearSVC handled imbalance best

After tuning and class-weighting, LinearSVC gave the most balanced performance across all classes.

Recommendations

1. Deploy the optimized LinearSVC model for continuous, automated sentiment tracking.
2. Pay closer attention to negative sentiment trends — they often signal developing customer issues.
3. Collect more negative tweet samples to strengthen representation of the minority class.
4. Explore advanced techniques such as SMOTE or transformer-based models (e.g. BERT) for further gains.
5. Regularly retrain the model on fresh social media data to adapt to evolving language and trends.

Conclusion

This project built and evaluated machine learning models to classify the sentiment of tweets about Apple and Google products. After thorough preprocessing, feature extraction, and model comparison, the optimized LinearSVC model performed best overall — achieving 90.3% accuracy and a Macro F1-score of ~0.76-0.78 with relatively balanced predictions across sentiment classes.

These results demonstrate that machine learning can effectively interpret social media sentiment, providing valuable insights for brand monitoring, customer feedback analysis, and strategic decision-making.

90.3%

Test Accuracy

~0.76-0.78

Macro F1-Score

Thank You